

Shifted-state intrinsic Hessian and nonzero-neighbourhood gap boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope: the first nonzero shifted-state neighbourhood

R-155 proves, at the zero-control origin of a pure dyadic fixed-law affine family and with $v = (H, G)$, the exact lower bound

$$D^2\mathcal{A}(0)[v, v] > \frac{147}{1000}(\|H\|_{\text{HS}}^2 + \|G\|_{\text{HS}}^2). \quad (1.1)$$

R-156 instantiates the smallest genuinely recursive predictable chart and extends (1.1), by finite-dimensional continuity, to some nonzero coefficient neighbourhood. Let ξ_p and ξ_{2p} be independent whitened both-sign raw roots and define

$$\chi(A, B) = (A\xi_p, B(\xi_{2p} + A\xi_p)). \quad (1.2)$$

The first component is revealed at $2p$ and the second at $4p$; hence

$$h_{2p} \in \mathcal{F}_p \quad \text{and} \quad h_{4p} \in \mathcal{F}_{2p}. \quad (1.3)$$

At the registered fixed finite cutoff, positive floor, covariance-matched common-even background and exact nonaliased torus integration, there exists some $\delta > 0$ such that

$$\|A\|_{\text{HS}} + \|B\|_{\text{HS}} < \delta \quad (1.4)$$

implies, for every nonzero coefficient direction $v = (H, G)$,

$$\boxed{D^2(\mathcal{A} \circ \chi)_{A,B}[v, v] > \frac{1}{10}(\|H\|_{\text{HS}}^2 + \|G\|_{\text{HS}}^2).} \quad (1.5)$$

The corresponding intrinsic physical-source Hessian, restricted to the tangent image of $D\chi_{A,B}$, also remains above $1/10$ in its induced source Gram metric after possibly decreasing δ .

The radius in (1.4) is existential. R-156 supplies no numerical value and no radius uniform in the cutoff, floor, refinement, chart, or realised Gaussian past. It does not prove global finite-amplitude convexity, arbitrary predictable nonlinear/revisit feedback, T-050, A13, Nelson, an interacting measure, any phase, lattice, vacuum, BCC, or PDE verdict, or Sector-A closure.

2. Exact shifted-state differential and source geometry

For directions $v = (H, G)$ and $w = (K, L)$, direct differentiation of (1.2) gives

$$\begin{aligned} k_v &= D\chi_{A,B}v = (H\xi_p, G\xi_{2p} + (GA + BH)\xi_p), \\ D^2\chi_{A,B}[v, w] &= (0, (GK + LH)\xi_p). \end{aligned} \quad (2.1)$$

The physical composition order is fixed. Exchanging the two Hessian arguments makes $GK + LH$ reappear; it does not replace either product by an illegitimate label-swapped composition.

Whitened independence gives the intrinsic tangent Gram

$$\boxed{\mathcal{G}_{A,B}(v, w) = \langle H, K \rangle_{\text{HS}} + \langle G, L \rangle_{\text{HS}} + \langle GA + BH, LA + BK \rangle_{\text{HS}}.} \quad (2.2)$$

In particular,

$$\mathcal{G}_{A,B}(v, v) = \|H\|_{\text{HS}}^2 + \|G\|_{\text{HS}}^2 + \|GA + BH\|_{\text{HS}}^2 > 0 \quad (2.3)$$

for $v \neq 0$. There is no extra low variable in (2.1); all three summands are coordinates of the same predictable source tangent.

The Cameron–Martin part of the scalar action is

$$\mathcal{C}(A, B) = \frac{9}{20} \left(\|A\|_{\text{HS}}^2 + \|B\|_{\text{HS}}^2 + \|BA\|_{\text{HS}}^2 \right). \quad (2.4)$$

Its controller-coordinate Hessian is

$$\boxed{D^2\mathcal{C}(A, B)[v, w] = \frac{9}{10} [\mathcal{G}_{A,B}(v, w) + \langle BA, GK + LH \rangle_{\text{HS}}].} \quad (2.5)$$

The second term in (2.5) is a gradient–chart-acceleration connection. It is absent from the intrinsic affine physical-source Hessian and vanishes at $A = B = 0$. Thus the positive 9/10 intrinsic source metric may not be copied unchanged into a nonlinear controller-coordinate Hessian away from the origin.

3. One production scalar, projected force, and all owners once

On a fixed predictable chart π , R-141 writes the complete production action as one scalar

$$\mathcal{A}_\pi(h) = \sum_{\alpha} \mathbb{E} \mathcal{P}_{\text{comp}, \alpha}(Z_h) + \frac{9}{20} \|h\|_{\mathcal{S}_\pi}^2 + \frac{3}{20} \mathbb{E} \|Z_{K,h}\|_6^6. \quad (3.1)$$

Its projected force and intrinsic source Hessian are

$$\begin{aligned} g_\pi^{\text{prod}} &= \frac{9}{10} h + P_{\text{pred}} L_\pi^* G_{\text{end}}, \\ H_\pi(h) &= \frac{9}{10} I + P_{\text{pred}} L_\pi^* (Q_{\text{comp}} + Q_6) L_\pi P_{\text{pred}}. \end{aligned} \quad (3.2)$$

Consequently the complete pullback Hessian is exactly

$$\boxed{D^2(\mathcal{A}_\pi \circ \chi)_{A,B}[v, w] = \langle k_v, H_\pi(h_{A,B}) k_w \rangle + \mathbb{E} \langle (g_\pi^{\text{prod}})_{4p}, (GK + LH) \xi_p \rangle.} \quad (3.3)$$

The second term is the unique controller-coordinate connection. It already contains the source, endpoint, sextic and trace first variations; adding them again as separate reserves would double count the same scalar action.

For the intrinsic affine physical interpolation

$$h = h_{A,B} + t k_v + s k_w, \quad (3.4)$$

there is no chart acceleration, and (3.3) reduces to

$$D^2\mathcal{A}_\pi(h_{A,B})[k_v, k_w] = \langle k_v, H_\pi(h_{A,B}) k_w \rangle. \quad (3.5)$$

The production current still has its genuine internal second jet; it is not controller curvature. In the R-153 conditional notation $W = m + X$, $n_i = \partial_i m$, $z = L_\pi k_v$, $w = L_\pi k_w$, $u_{i,z} = \partial_i z$, the A7 block is

$$K_E[z, w] = \int \sum_i \left[u_{i,z}^T \overline{B} u_{i,w} + u_{i,z}^T \overline{DB}[w], n_i + u_{i,w}^T \overline{DB}[z], n_i + \frac{1}{2} \text{Tr} \left(\overline{D^2 B}[z, w] (n_i n_i^T - \Gamma_{<,i}) \right) \right] dx. \quad (3.6)$$

The sixth-power contribution is the single global form

$$K_6[z, w] = \frac{9}{10} \int \mathbb{E} \left[|W|^4 z \cdot w + 4|W|^2 (W \cdot z)(W \cdot w) \right] dx, \quad (3.7)$$

which is positive semidefinite on the diagonal. Equations (3.2), (3.6), and (3.7) are one source payment, one complete endpoint/current owner, and one sextic owner. Forward and reverse entries are adjoints of the same bilinear Hessian.

4. Two-block acceptance and why no new low is allowed

For the two affine source blocks, write the complete intrinsic Hessian and source metric as M and $\mathcal{G}$. The target is

$$N = M - \frac{1}{10} \mathcal{G} \succeq 0 \quad \text{and} \quad N = \begin{pmatrix} E & C \\ C^* & F \end{pmatrix}. \quad (4.1)$$

The forward block is C ; its sole legal reverse is C^* . The balanced term is the real symmetric cross $2 \text{Re} \langle x, Cy \rangle$, not a third reserve.

If $F > 0$, exact block congruence gives

$$N \succeq 0 \iff F > 0 \quad \text{and} \quad E - CF^{-1}C^* \succeq 0. \quad (4.2)$$

For semidefinite F , (4.2) requires the Douglas range condition and a reduced solution of $F^{1/2}X = C^*$. A free Schur minimisation is legitimate only if the lower coordinate is an actual independent variable and its stationary minimiser exists. No such variable occurs in (1.2). The R-141 source-Doob low component is an isometric reparameterisation of h , while historical mean, forest, future-variance and returned-low expressions are owner decompositions already contained in Q_{comp} . R-156 therefore adds no independent low/Feshbach block.

5. Continuation proof and the exact global boundary

At fixed cutoff and positive floor, (3.1) is a smooth function on a finite dimensional coefficient space, and (1.2) is polynomial. Hence both

$$(A, B) \mapsto D^2(A \circ \chi)_{A,B}, \quad (A, B) \mapsto (D\chi_{A,B}^* H_\pi D\chi_{A,B}, \mathcal{G}_{A,B}) \quad (5.1)$$

are continuous. At $A = B = 0$, the projected-force connection in (3.3) vanishes by the R-155 stationary translation/Fourier cancellation, and $\mathcal{G}_{0,0} = I$. The exact headroom over the desired target is

$$\frac{147}{1000} - \frac{1}{10} = \frac{47}{1000}. \quad (5.2)$$

Continuity in operator norm therefore supplies a ball on which the change is, for example, less than half of (5.2). On that ball the retained lower bound is at least

$$\frac{147}{1000} - \frac{47}{2000} = \frac{247}{2000} > \frac{1}{10}. \quad (5.3)$$

This proves (1.5), after shrinking once more for the intrinsic generalised eigenvalue statement.

The argument cannot be applied to the general factor-three R-155 family: its registered origin gap is only

$$\frac{7}{250} < \frac{1}{10}. \quad (5.4)$$

Nor does compactness produce a usable uniform radius when the cutoff, floor, chart, or dimension changes. R-153 separately shows why a deterministic fibrewise collar cannot be uniform over an unbounded non-degenerate Gaussian past.

6. Nonlinear pullback curvature no-go

Take an aligned rank-one projection with $\|H\|_{\text{HS}}^2 = \|G\|_{\text{HS}}^2 = \|GH\|_{\text{HS}}^2 = 1$, and put $A = tH$, $B = sG$. Then

$$\mathcal{C}(t, s) = \frac{9}{20}(t^2 + s^2 + t^2 s^2), \quad (6.1)$$

$$D^2\mathcal{C} = \frac{9}{10} \begin{pmatrix} 1 + s^2 & 2ts \\ 2ts & 1 + t^2 \end{pmatrix}. \quad (6.2)$$

On $t = s = R$, the tangent source Gram and controller Hessian have eigenvalues

$$\text{spec } \mathcal{G}_R = \{1, 1 + 2R^2\}, \quad \text{spec } D^2\mathcal{C}_R = \left\{ \frac{9}{10}(1 - R^2), \frac{9}{10}(1 + 3R^2) \right\}. \quad (6.3)$$

At $R = 2$, these are respectively $\{1, 9\}$ and $\{-27/10, 117/10\}$. Thus even a convex Cameron–Martin norm becomes indefinite in large nonlinear controller coordinates. This is not a counterexample to intrinsic production convexity or to T-050. It falsifies only the route that treats nonlinear pullback curvature as if it were the intrinsic 9/10 source reserve.

7. Devil’s-advocate review

1. **Objection: continuity is being used to claim a global nonlinear theorem. UPHOLD AS A BOUNDARY.** The theorem is existential and local at one fixed cutoff, floor and two-stage chart. No numerical or uniform radius is claimed.
2. **Objection: the negative eigenvalue in Section 6 disproves production convexity. DISMISSED.** It is exactly the gradient–chart–acceleration term in (2.5). The intrinsic source metric in (2.3) remains positive definite.
3. **Objection: the source, endpoint, sextic and connection may be estimated as four independent reserves. DISMISSED.** They are derivatives of the one scalar (3.1). Equation (3.3) owns the full connection once, and (3.6)–(3.7) own the intrinsic endpoint and sextic forms once.
4. **Objection: balanced and legal reverse are two new payments. DISMISSED.** They are the cross term and adjoint orientation of the same self-adjoint Hessian block.
5. **Objection: a low/Feshbach variable can be inferred from prior owner names. UPHOLD AS PROHIBITED.** No independent low coordinate is defined in the production chart. A Schur complement may analyse the two actual source blocks but may not invent a new reserve.
6. **Objection: the result supports BCC or the current PDE. UPHOLD AS PROHIBITED.** R-156 is phase-neutral local Hessian mathematics. It neither favours nor rejects BCC, a uniform state, another modulated state, another functional, or another equation.

8. Reproduction, route decision, and Result footer

From the repository root, run

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/  
a13_classii_shifted_state_nonzero_neighborhood_gap_boundary.py  
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/  
a13_classii_shifted_state_nonzero_neighborhood_gap_boundary_independent.py  
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/  
a13_classii_shifted_state_nonzero_neighborhood_gap_boundary_verify.py
```

R-156 is the minimum nonlinear/revisit successor requested by R-155. It establishes a real nonzero neighbourhood, identifies the complete intrinsic two-block target, and retires both nonlinear-coordinate source convexity and an invented independent-low completion. The remaining T-050 question is quantitative and uniform: obtain a production bound on the intrinsic operator over the required predictable charts, or produce a genuine counterexample. In parallel, the phase-neutral T-054/T-052 model tournament should reject unsupported candidate equations early. Truth, not a preferred phase or PDE, remains the selection rule.

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Result ID: A13-CLASSII-SHIFTED-STATE-NONZERO-NEIGHBORHOOD-GAP-  
BOUNDARY (R-156).  
Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.  
Precise statement: The pure-dyadic shifted-state chart has an existential  
nonzero fixed-cutoff coefficient neighbourhood with Hessian gap >1/10.  
Scope: Fixed cutoff/floor, exact torus, two-stage predictable shifted-state  
chart, intrinsic and pullback Hessians separated, existential radius.  
Dependencies: A1, A7, R-131, R-141, R-153, R-155, and Sections 1--5.  
Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.  
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe  
codes/foundations/a13_classii_shifted_state_nonzero_neighborhood_gap_  
boundary_verify.py  
Expected output: Integrated verifier PASS; primary 29/29; independent 21/21.  
Falsification gate: Any chart differential, source geometry, one-scalar  
chain rule, exact continuity margin, authority, record, or PDF gate fails.  
Tier before / after: T4 / T4; no promotion.  
No-overclaim statement: No numerical/uniform radius, global nonlinear/revisit,  
T-050/A13, Nelson/measure, phase/PDE, or Sector-A result follows.  
Next required action: Quantitative intrinsic operator test plus T-054/T-052.
```